第九次作业答案

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2024 年 5 月 17 日

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1. 计算二重积分:

$$\iint_D \left(x^2+y^2\right) \mathrm{d}x\,\mathrm{d}y$$

其中 D 是由曲线 $x^2-y^2=1, x^2-y^2=2, xy=1$ 和 $xy=2$ 围成的图形在第一象限中的那部分.

解. 取

$$(u,v)=\varphi(x,y)=\left(x^2-y^2,xy\right)$$
$$\Longrightarrow \det \mathbf{J}\varphi(x,y)=\begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix}=2\left(x^2+y^2\right),$$

不难验证在定义域上 φ 为微分同胚, 从而根据多重积分的变量代换有

$$\iint_D \left(x^2+y^2\right) \mathrm{d}x\,\mathrm{d}y=\iint_{\varphi(D)} \frac{x^2+y^2}{|\det \mathbf{J}\varphi(x,y)|}\mathrm{d}u\,\mathrm{d}v=\frac{1}{2}\int_1^2 \mathrm{d}u\int_1^2 \mathrm{d}v=\frac{1}{2}.$$

2. 计算由下列曲线围成的图形的面积:

(1). $(a_1x+b_1y+c_1)^2+(a_2x+b_2y+c_2)^2=1$, 其中 $a_1b_2\neq a_2b_1$;

(2). $\sqrt{x}+\sqrt{y}=\sqrt{a}, x=0$ 和 $y=0$.

解.

(1). 取

$$(u,v)=\varphi(x,y)=(a_1x+b_1y+c_1,a_2x+b_2y+c_2)$$
$$\Longrightarrow \det \mathbf{J}\varphi(x,y)=\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}=a_1b_2-a_2b_1$$

不难验证 φ 为微分同胚, 从而根据多重积分的变量代换有

$$\iint_D \mathrm{d}x\,\mathrm{d}y=\iint_{\varphi(D)} \frac{\mathrm{d}u\,\mathrm{d}v}{|\det \mathbf{J}\varphi(x,y)|}$$

由于 $\varphi(D):=\left\{(u,v)\mid u^2+v^2\leq 1\right\}$, 故

$$\iint_{\varphi(D)} \mathrm{d}u\,\mathrm{d}v=\pi \Longrightarrow \iint_D \mathrm{d}x\,\mathrm{d}y=\frac{\pi}{|a_1b_2-a_2b_1|}.$$

(2). 令 $u,v\geq 0$ 满足

$$(x,y)=\varphi(u,v)=\left(u^2,v^2\right)$$
$$\Longrightarrow \det \mathbf{J}\varphi(u,v)=\begin{vmatrix} 2u & 0 \\ 0 & 2v \end{vmatrix}=4uv.$$

不难验证 φ 在除去 $xy=0$ 的一个零测集上为微分同胚, 从而根据多重积分的变量代换的推论有

$$\iint_D \mathrm{d}x\,\mathrm{d}y=\iint_{\varphi^{-1}(D)} 4uv\,\mathrm{d}u\,\mathrm{d}v=\int_0^{\sqrt{a}} \mathrm{d}u\int_0^{\sqrt{a}-u} 4uv\,\mathrm{d}v=\frac{1}{6}a^2.$$

3. 求证:

$$\iint_{|x|+|y|\leqslant 1} f(x+y)\mathrm{d}x\,\mathrm{d}y = \int_{-1}^{1} f(t)\mathrm{d}t.$$

证明. 取

$$\begin{aligned}(u,v) = \varphi(x,y) &= (x+y, x-y)\\ \Longrightarrow \det \mathbf{J}\varphi(x,y) &= \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = 2,\end{aligned}$$

不难验证在定义域上 φ 为微分同胚, 且注意到 $|x|+|y| = \max(|u|,|v|)$, 从而

$$\begin{aligned}\iint_{|x|+|y|\leq 1} f(x+y)\mathrm{d}x\,\mathrm{d}y &= \iint_{|u|,|v|\leq 1} \frac{f(u)}{|\det \mathbf{J}\varphi(x,y)|}\mathrm{d}u\,\mathrm{d}v\\ &= \frac{1}{2}\int_{-1}^{1} f(u)\mathrm{d}u \int_{-1}^{1} f(v)\mathrm{d}v\\ &= \int_{-1}^{1} f(u)\mathrm{d}u\end{aligned}$$

□

4. 设 D 是由曲线 $xy = 1, xy = 2, y = x$ 和 $y = 4x$ 围成的图形在第一象限中的那一部分. 求证:

$$\iint_D f(xy)\mathrm{d}x\,\mathrm{d}y = \ln 2 \int_1^2 f(t)\mathrm{d}t.$$

证明. 取

$$\begin{aligned}(u,v) = \varphi(x,y) &= \left(xy, \frac{y}{x}\right)\\ \Longrightarrow \det \mathbf{J}\varphi(x,y) &= \begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} = \frac{2y}{x} = 2v.\end{aligned}$$

不难验证在定义域上 φ 为微分同胚, 从而

$$\iint_D f(xy)\mathrm{d}x\,\mathrm{d}y = \int_1^2 f(u)\mathrm{d}u \int_1^4 \frac{1}{2v}\,\mathrm{d}v = \ln 2 \int_0^1 f(u)\mathrm{d}u.$$

□

5. 计算下列二重积分:

(1). $\iint_{\pi^2\leqslant x^2+y^2\leqslant 4\pi^2} \sin\sqrt{x^2+y^2}\,\mathrm{d}x\,\mathrm{d}y$;

(2). $\iint\limits_{\substack{x^2+y^2\leqslant 1\\ x\leqslant 0, y\leqslant 0}} \arctan\frac{y}{x}\,\mathrm{d}x\,\mathrm{d}y$;

解.

(1). 做极坐标变换

$$x = r\cos\theta, \quad y = r\sin\theta$$

则

$$\begin{aligned}\iint_{\pi^2\leq x^2+y^2\leq 4\pi^2}\sin\sqrt{x^2+y^2}\,\mathrm{d}x\,\mathrm{d}y &= \int_{\pi}^{2\pi} r\sin r\,\mathrm{d}r\int_0^{2\pi}\mathrm{d}\theta\\ &= 2\pi\int_{\pi}^{2\pi}-r\,\mathrm{d}(\cos r)\\ &= 2\pi\left[-r\cos r|_{\pi}^{2\pi}+\int_{\pi}^{2\pi}\cos r\,\mathrm{d}r\right]=-6\pi^2\end{aligned}$$

(2). 做极坐标变换

$$x=r\cos\theta,\quad y=r\sin\theta$$

则

$$\iint_D \arctan\frac{y}{x}\,\mathrm{d}x\,\mathrm{d}y=\int_0^1 r\,\mathrm{d}r\int_{\pi}^{\frac{3}{2}\pi}(\theta-\pi)\mathrm{d}\theta=\frac{\pi^2}{16}.$$

6. 设常数 $a,b>0$,

$$D=\left\{(x,y):\frac{x^2}{a^2}+\frac{y^2}{b^2}\leqslant 1\text{ 且}x\geqslant y\geqslant 0\right\}.$$

计算二重积分

$$\iint_D\sqrt{\frac{x^2}{a^2}+\frac{y^2}{b^2}}\,\mathrm{d}x\,\mathrm{d}y$$

解. 做变换

$$\begin{aligned}&(x,y)=\varphi(r,\theta)=(ar\cos\theta,br\sin\theta)\\ \Longrightarrow\ &\det\mathbf{J}\varphi(r,\theta)=\begin{vmatrix}a\cos\theta & -ar\sin\theta\\ b\sin\theta & br\cos\theta\end{vmatrix}=abr,\end{aligned}$$

从而由多重积分的变量代换的推论有

$$\begin{aligned}\iint_D\sqrt{\frac{x^2}{a^2}+\frac{y^2}{b^2}}\,\mathrm{d}x\,\mathrm{d}y &= \int_0^1 abr\,\mathrm{d}r\int_0^{\arctan\frac{a}{b}}\sqrt{\frac{a^2r^2\cos^2\theta}{a^2}+\frac{b^2r^2\sin\theta}{b^2}}\,\mathrm{d}\theta\\ &= ab\int_0^1 r^2\,\mathrm{d}r\int_0^{\arctan\frac{a}{b}}\mathrm{d}\theta=\frac{1}{3}ab\arctan\frac{a}{b}\end{aligned}$$

7. 证明:

$$\iint_{[0,1]^2}(xy)^{xy}\,\mathrm{d}x\,\mathrm{d}y=\int_0^1 t^t\,\mathrm{d}t$$

证明. 做换元

$$\begin{aligned}&(u,v)=\varphi(x,y)=\left(xy,\frac{y}{x}\right)\\ \Longrightarrow\ &\det\mathbf{J}\varphi(x,y)=\begin{vmatrix}y & x\\ -\frac{y}{x^2} & \frac{1}{x}\end{vmatrix}=\frac{2y}{x}=2v.\end{aligned}$$

从而新的积分区域变为 $\left\{(u,v) \mid u \in (0,1], v \in \left[u, \frac{1}{u}\right]\right\}$ 由多重积分的变量代换的推论有

$$\iint_{[0,1]^2} (xy)^{xy}\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 u^u\,\mathrm{d}u \int_u^{1/u} \frac{\mathrm{d}v}{2v} = -\int_0^1 u^u \ln u\,\mathrm{d}u.$$

注意到

$$\mathrm{d}\left(u^u\right) = u^u(1+\ln u)\mathrm{d}u = \left(u^u + u^u \ln u\right)\mathrm{d}u$$

从而

$$-\int_0^1 u^u \ln u\,\mathrm{d}u = \int_0^1 u^u\,\mathrm{d}u - \int_0^1 \mathrm{d}\left(u^u\right) = \int_0^1 u^u\,\mathrm{d}u$$

□

8. 设常数 a, b 不全为 0 . 求证:

$$\iint_{x^2+y^2\leqslant 1} f(ax+by+c)\mathrm{d}x\,\mathrm{d}y = 2\int_{-1}^1 \sqrt{1-t^2} f\left(t\sqrt{a^2+b^2}+c\right)\mathrm{d}t.$$

证明. 做换元

$$\begin{aligned}(u,v) = \varphi(x,y) &= \frac{1}{\sqrt{a^2+b^2}}(ax+by, ay-bx)\\ \Longrightarrow \det \mathbf{J}\varphi(x,y) &= \frac{1}{\sqrt{a^2+b^2}}\begin{vmatrix} a & b \\ -b & a \end{vmatrix} = 1.\end{aligned}$$

从而新的积分区域变为 $u^2+v^2 \leq 1$. 由多重积分的变量代换的推论有

$$\begin{aligned}\iint_{x^2+y^2\leq 1} f(ax+by+c)\mathrm{d}x\,\mathrm{d}y &= \iint_{u^2+v^2\leq 1} f\left(\sqrt{a^2+b^2}u+c\right)\mathrm{d}u\,\mathrm{d}v\\ &= \int_0^1 f\left(\sqrt{a^2+b^2}u+c\right)\mathrm{d}u \int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} \mathrm{d}v\\ &= 2\int_0^1 \sqrt{1-u^2} f\left(\sqrt{a^2+b^2}u+c\right)\mathrm{d}u\end{aligned}$$

□

9. 设 f 是单变量函数，且连续可导. 令

$$F(t) = \iint_{[0,t]^2} f(xy)\mathrm{d}x\,\mathrm{d}y$$

求证:

(1). $F'(t) = \frac{2}{t}\left(F(t) + \iint_{[0,t]^2} xyf'(xy)\mathrm{d}x\,\mathrm{d}y\right)$;

(2). $F'(t) = \frac{2}{t}\int_0^{t^2} f(s)\mathrm{d}s$.

证明.

(1). 做换元 $x=tu, y=tv$, 则

$$F(t)=\iint_{[0,t]^2} f(xy)\mathrm{d}x\,\mathrm{d}y=\iint_{[0,1]^2} t^2 f\left(t^2uv\right)\mathrm{d}u\,\mathrm{d}v,$$

从而

$$F'(t)=\iint_{[0,1]^2} 2tf\left(t^2uv\right)+2t^3f'\left(t^2uv\right)\mathrm{d}u\,\mathrm{d}v.$$

重新做换元 $x=\frac{u}{t}, y=\frac{v}{t}$, 则

$$\begin{aligned}F'(t)&=\frac{2}{t}\iint_{[0,t]^2} f(xy)+xyf'(xy)\mathrm{d}x\,\mathrm{d}y\\&=\frac{2}{t}\left(F(t)+\iint_{[0,t]^2} xyf'(xy)\mathrm{d}x\,\mathrm{d}y\right).\end{aligned}$$

(2). 根据对称性,

$$F(t)=2\int_0^t \mathrm{d}x\int_0^x f(xy)\mathrm{d}y\Longrightarrow F'(t)=2\int_0^t f(ty)\mathrm{d}y.$$

做换元 $s=ty$ 就有

$$F'(t)=\frac{2}{t}\int_0^{t^2} f(s)\mathrm{d}s$$

□

10. 计算二重积分:

$$\iint_D \frac{\mathrm{d}x\,\mathrm{d}y}{xy\left(\ln^2 x+\ln^2 y\right)}$$

其中 D 是由在第一象限里的圆周 $x^2+y^2=1$ 与直线 $x+y=1$ 所围成的图形.

证明. 做变量代换

$$(u,v)=\varphi(x,y)=(\ln x,\ln y)\Longrightarrow \det \mathbf{J}\varphi(x,y)=\frac{1}{xy}.$$

从而

$$\iint_D \frac{\mathrm{d}x\,\mathrm{d}y}{xy\left(\ln^2 x+\ln^2 y\right)}=\iint_{\varphi(D)}\frac{\mathrm{d}u\,\mathrm{d}v}{u^2+v^2}.$$

其中

$$\varphi(D):=\left\{(u,v)\mid e^{2u}+e^{2v}\le 1\le e^u+e^v\right\}.$$

再做极坐标变换 $u=r\cos\theta, v=r\sin\theta$, 则

$$\iint_{\varphi(D)}\frac{\mathrm{d}u\,\mathrm{d}v}{u^2+v^2}=\iint_{D'}\frac{r\,\mathrm{d}r\,\mathrm{d}\theta}{r^2}=\iint_{D'}\frac{\mathrm{d}r\,\mathrm{d}\theta}{r},$$

其中

$$D' := \left\{ (r,\theta) \mid e^{2r\cos\theta} + e^{2r\sin\theta} \le 1 \le e^{r\cos\theta} + e^{r\sin\theta} \right\}$$

此时对一个确定的 θ, 记对应的区域为 $r \in [\varphi_1(\theta), \varphi_2(\theta)]$, 则注意到有 $\varphi_2(\theta) = 2\varphi_1(\theta)$, 从而

$$\iint_{D'} \frac{\mathrm{d}r\,\mathrm{d}\theta}{r} = \int_{\pi}^{\frac{3}{2}\pi} \mathrm{d}\theta \int_{\varphi_1(\theta)}^{2\varphi_1(\theta)} \frac{\mathrm{d}r}{r} = \int_{\pi}^{\frac{3}{2}\pi} \ln 2\,\mathrm{d}\theta = \frac{\pi}{2}\ln 2.$$

□